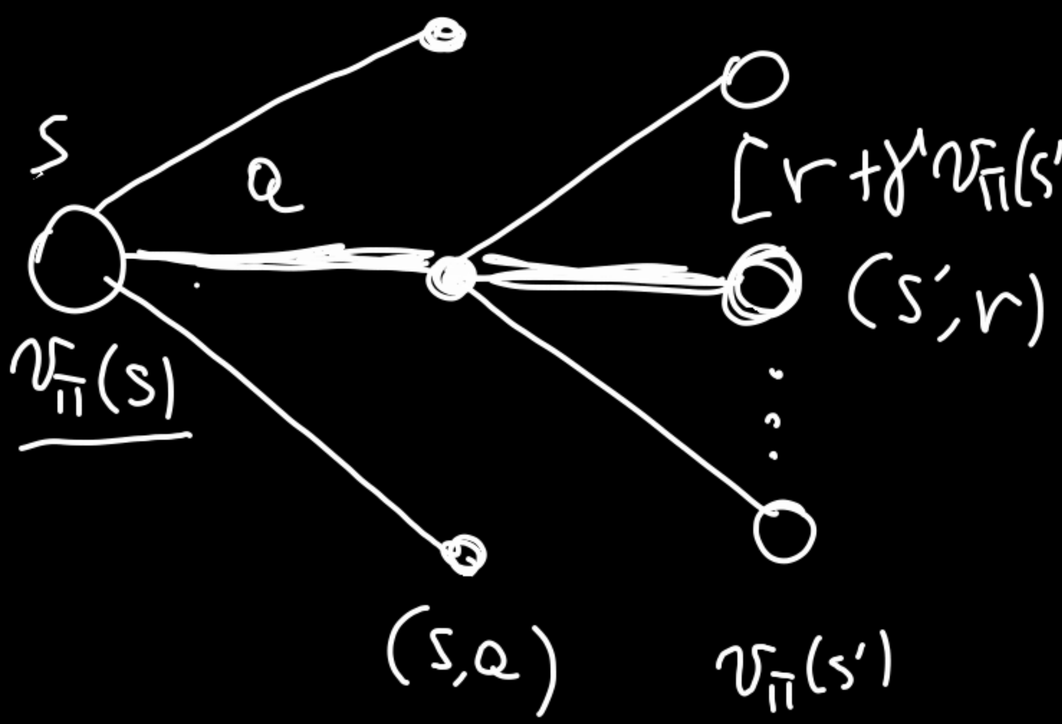
5/11/26 – DYNAMIC PROGRAMMING METHODS + INTRO. TO MONTECARLO METHODS

p IS KNOWN

BELLMAN EQ. FOR $v_\pi(s)$



$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a)\left[r+\gamma v_\pi(s')\right] \quad \forall s$$

MODEL OF ENV.

SET OF $|S|$ LINEAR EQ. IN $|S|$ VARIABLES $(v_\pi(s_1), v_\pi(s_2), \dots, v_\pi(s_{|S|}))$

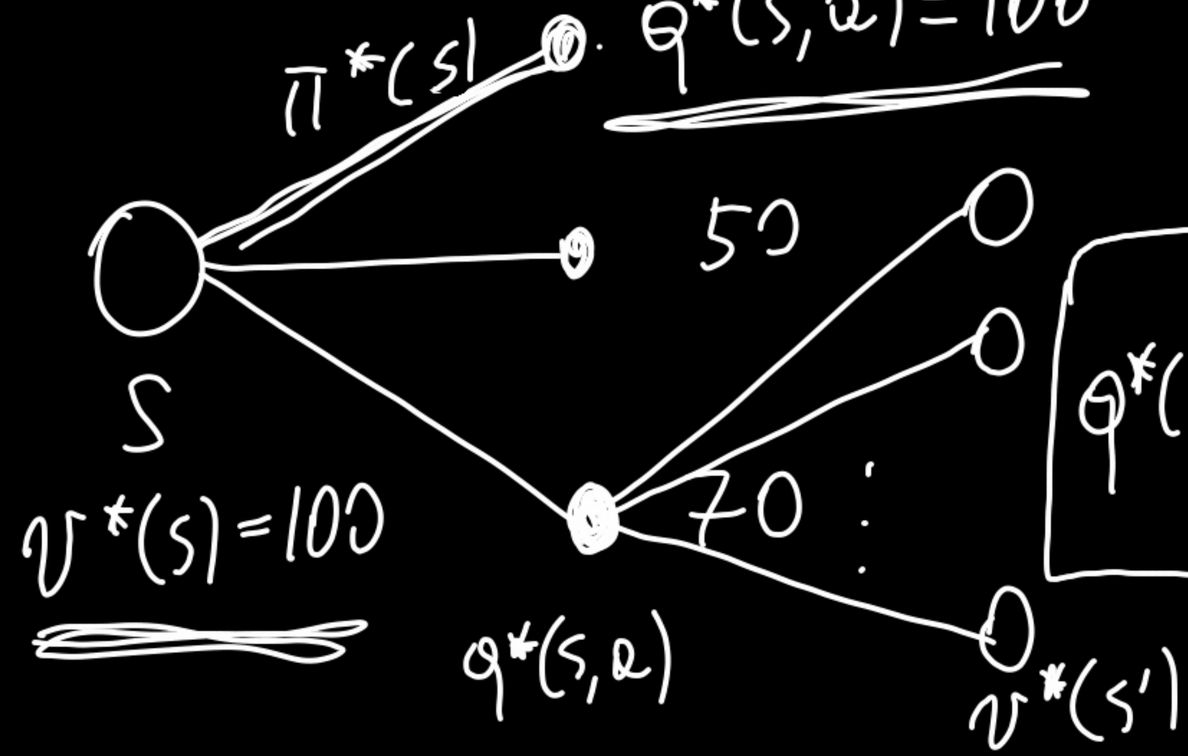
NO KNOWLEDGE OF ENVIRONMENT (LEARNING)

OPTIMAL POLICIES

$$\pi^* \geq \pi \;, \forall \pi \quad \left(v_{\pi^*}(s) \geq v_\pi(s), \forall s, \pi \right)$$

$$\pi^*(s) = \arg\max_a \left(q^*(s,a) \right)$$

THE OPTIMAL POLICY IS GREEDY WRT THE q^* FUNCTION.



$$q^*(s,a) = \sum_{s',r} p(s',r|s,a)\left[r + \right.$$

$$v^*(s) = \max_a \{ q^*(s,a) \} = \max_a \left\{ \sum_{s',r} p(s',r|s,a)[r + \gamma v^*(s')] \right\} \quad \forall s$$

- $|S|$ NONLINEAR EQ. IN $|S|$ VARIABLES (THE VALUES).
- IT HAS A UNIQUE SOLUTION ($v^*(s)$, $\forall s$).
- IN MOST OF THE PRACTICAL CASES IT CANNOT BE SOLVED (NONLINEAR, HUGE STATE SPACES).

$\gamma v^*(s')]$

B.E. FOR $q^*(s,a)$ $\forall s, a$

OPTIMAL $q^*, v^* \longrightarrow \pi^*$ → Q-LEARNING

$$\pi^*(s) = \arg\max_a \{ q^*(s,a) \}, \forall s \quad \text{(DETERMINISTIC)}.$$

$$\pi^*(s) = \arg\max_a \left\{ \sum_{s',r} p(s',r|s,a)\left[r + \gamma v^*(s') \right] \right\}$$

↓

SIMILAR TO MPC ($N=1$), WE NEED THE MODEL.

DYNAMIC PROGRAMMING METHODS.

(MODEL OF ENV. IS KNOWN).

PROBLEM OF POLICY EVALUATION

GIVEN A $\pi(s,a)$, HOW TO COMPUTE $v_\pi(s)$?

(. ~~WRITE THE SYSTEM OF B.E. AND SOLVE IT~~) .

$\pi' \geq \pi$

$v_{\pi'}(s) \geq v_\pi(s), \forall s$

(ITERATIVE) POLICY EVALUATION (COMPUTATION OF $v_\pi(s)$)

$\pi \longrightarrow (v_\pi)$

B.E.

BELLMAN OPERATOR

$$v_\pi(s) = \sum_a \pi(a|s) \cdot \sum_{s',r} p(s',r|s,a)\left[r + \gamma v_\pi(s')\right], \forall s.$$

$v_\pi^k(s)$: k-TH ESTIMATE OF $v_\pi(s)$.

$v_\pi^0(s)$ INITIAL ESTIMATE/GUESS FOR $v_\pi(s)$.

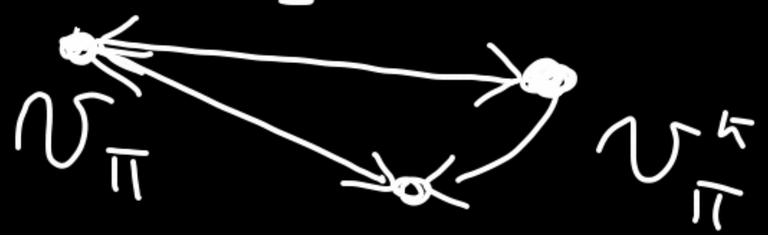
PICARD ITERATION

$$v_\pi^{k+1}(s) = \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a)\left[r + \gamma v_\pi^k(s')\right], \forall s.$$

IT IS POSSIBLE TO PROVE THAT, AS $k \to \infty$, $v_\pi^k(s) \to v_\pi(s)$

(FIXED-POINT THEOREM, CONTRACTION THEORY)

$$\max_s \left(v_\pi(s) - v^k(s) \right)$$



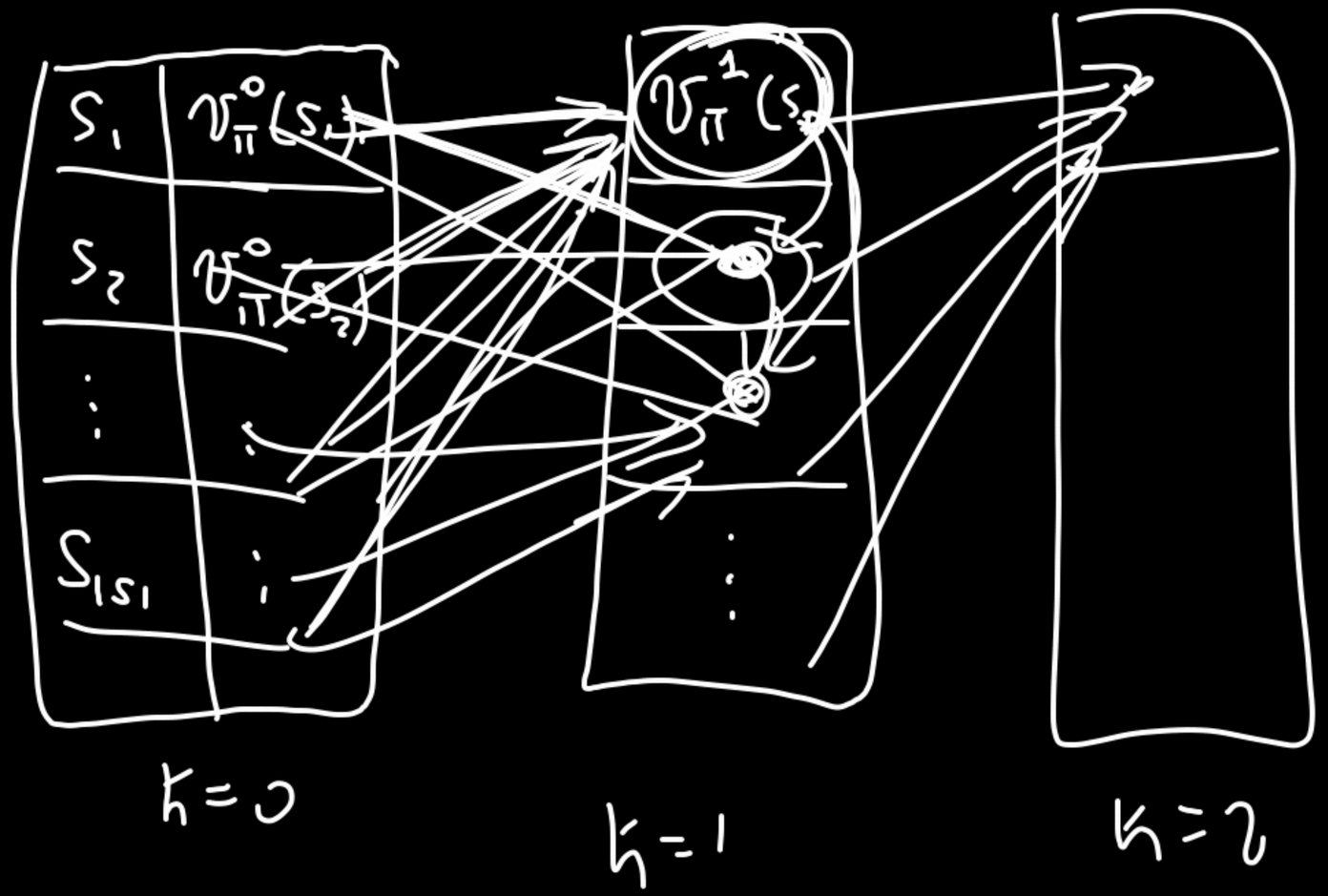
IN THEORY, ITERATIVE POLICY ITERATION NEEDS INFINITE-MANY ITERATIONS.

→ EXPECTED UPDATE

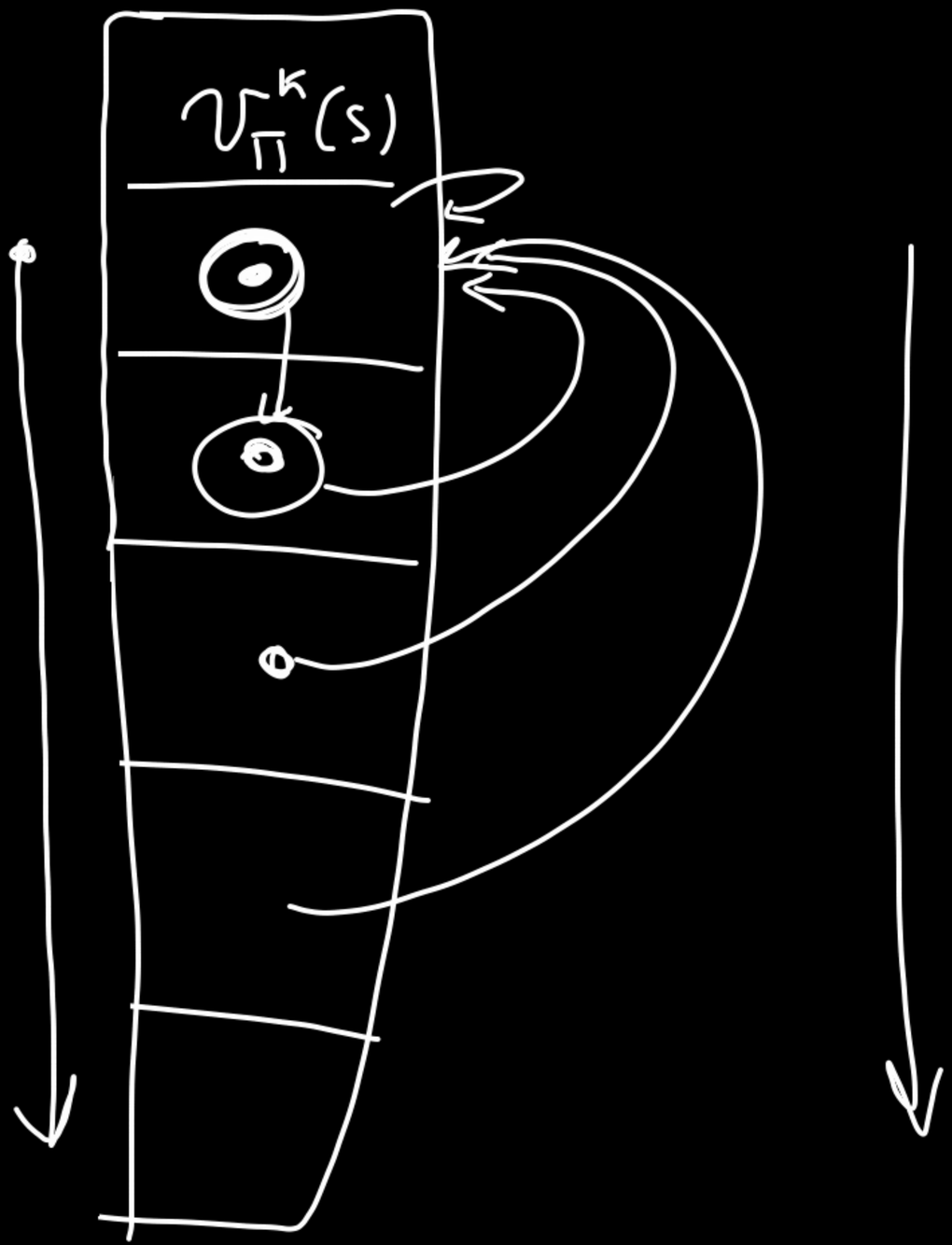
BOOTSTRAPPING METHOD

ITERATION BY:

- SEQUENTIAL FULL SWEEP OF STATE SPACE S.



IN-PLACE UPDATE OF $v_\pi^k(s)$



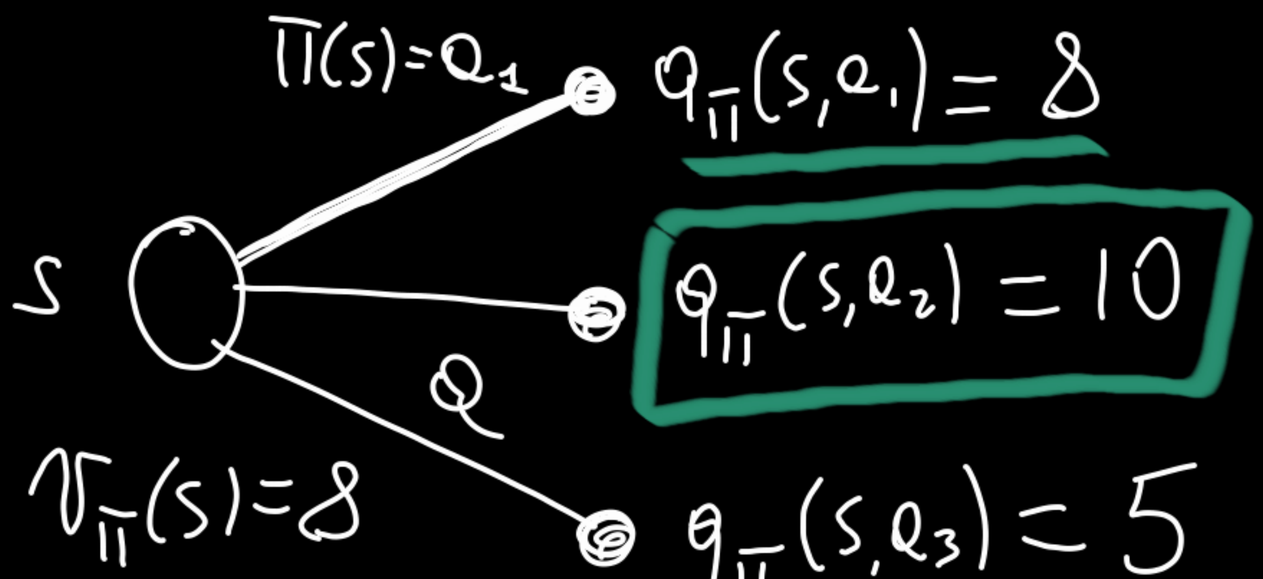
HOW TO TACKLE LARGE STATE SPACES?

(FOCUSED UPDATES)

$\pi \xrightarrow{E} \begin{pmatrix} v_\pi(s) \\ q_\pi(s,a) \end{pmatrix} \xrightarrow{I} \pi' \quad (\pi' \geq \pi)$

I - POLICY IMPROVEMENT - FIND A BETTER POLICY

DETERMINISTIC POLICIES ($\pi(s)$)



$v_\pi(s)$ $\pi(s)$ ACTION SELECTED IN s BY π.

HOW GOOD IS TO SELECT IN s ANOTHER GENERIC ACTION $a \neq \pi(s)$?

WE CAN ANSWER BY COMPARING $v_\pi(s)$ AND

$$q_\pi(s,a) = \sum_{s',r} p(s',r|s,a)\left[r + \gamma v_\pi(s')\right]$$

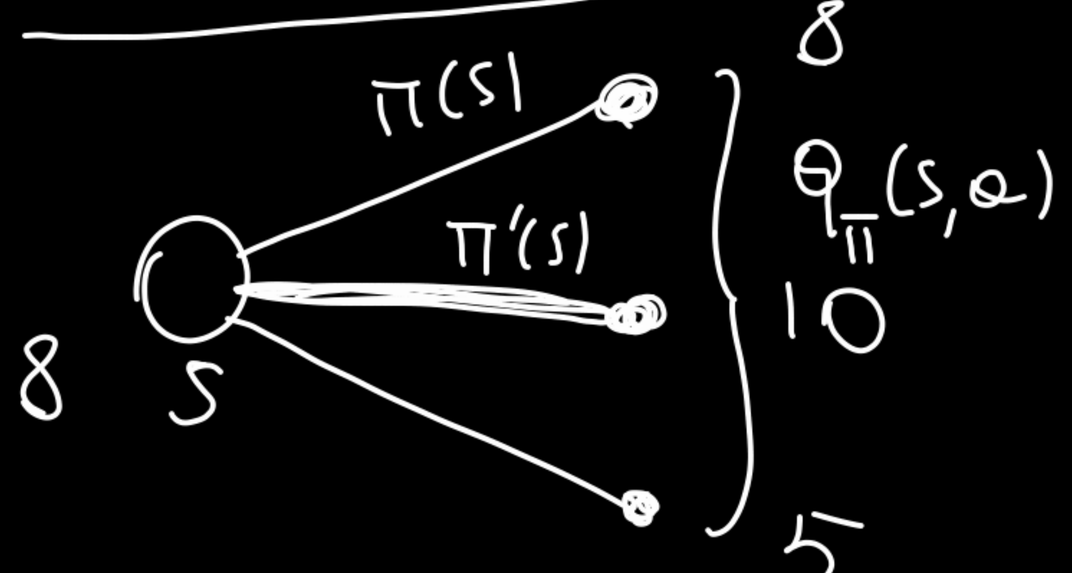
IF $q_\pi(s,a) > v_\pi(s)$, THEN THE NEW POLICY π' THAT IN s PICKS a INSTEAD OF $\pi(s)$ IS BETTER.

POLICY IMPROVEMENT THEOREM:

GIVEN POLICIES π' AN π', IF:

$$q_\pi(s, \pi'(s)) \geq v_\pi(s) \quad \forall s, \text{ THEN } \pi' \geq \pi.$$

POLICY IMPROVEMENT



THE IMPROVED POLICY π' IS THE GREEDY POLICY W.R.T $q_\pi(s,a)$

$\forall s$

$$\pi'(s) = \underset{a}{\operatorname{argmax}} \{ q_\pi(s,a) \}$$

BY DEFINITION

$$q_\pi(s, \pi'(s)) \geq v_\pi(s) \quad \forall s.$$

(POLICY IMPROVEMENT)

$\pi' \geq \pi$ $\left(v_{\pi'}(s) \geq v_\pi(s), \forall s \right)$

π' BETTER OR EQUALLY GOOD.

POLICY ITERATION (POLICY EVALUATION + POLICY IMPROVEMENT)

$$\pi_0 \xrightarrow{E} v_{\pi_0} \xrightarrow{I} \pi_1 \xrightarrow{E} v_{\pi_1} \cdots \longrightarrow \pi^*$$

E (ITERATIVE POLICY EVALUATION – ∞-MANY STEPS)

IN FINITE MDP, WE HAVE A FINITE NUMBER OF POSSIBLE DETERMINISTIC POLICIES (HOW MANY?)

$v_{\pi'}(s) \geq v_{\pi}(s)$, $\forall s$. AT SOME POINT, WE FIND THAT THE IMPROVED POLICY IS AS GOOD AS THE PREVIOUS ONE ($v_{\pi'}(s) = v_{\pi}(s), \forall s$)

WHAT CAN WE SAY WHEN $v_{\pi'}(s) = v_{\pi}(s), \forall s$?

(π' IS THE POLICY IMPROVING π).

$$\pi'(s) = \underset{a}{\operatorname{argmax}} \{ q_{\pi}(s,a) \}, \forall s$$

POLICY IMPROVEMENT ($\pi' \geq \pi$)

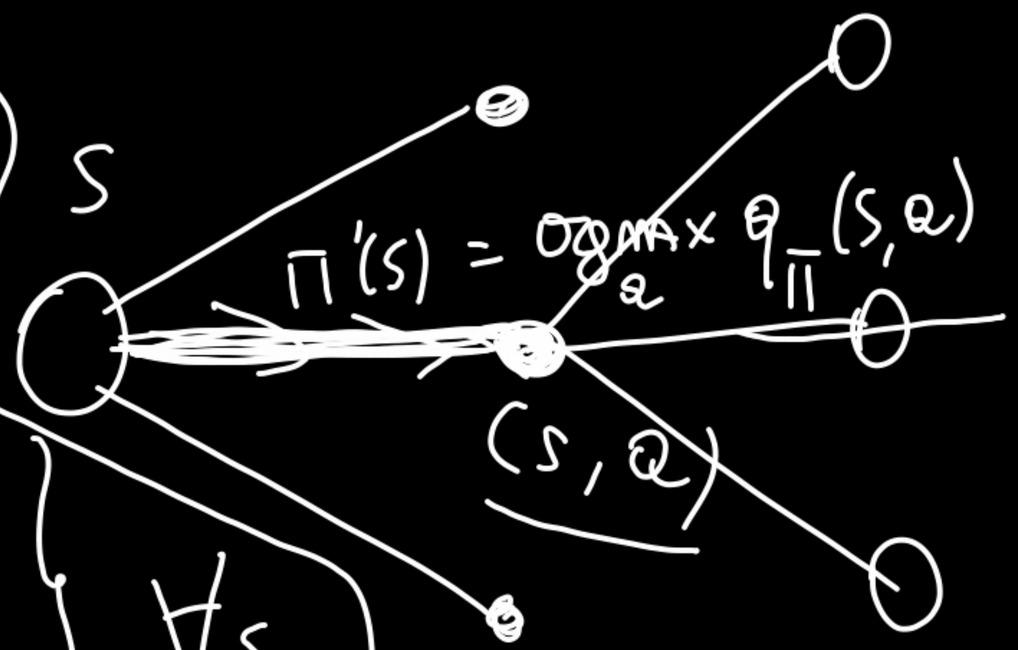
(THE IMPROVED POLICY IS GREEDY W.R.T THE q OF THE PREVIOUS POLICY).

$$\pi'(s) = \arg\max_a \{ q_\pi(s,a) \}, \forall s.$$

ASSUME $v_{\pi'}(s) = v_\pi(s)$, $\forall s$. WHAT DOES THIS MEAN ? $v_\pi(s')$

$$v_{\pi'}(s) = \max_a \{ q_\pi(s,a) \} =$$

$$= \max_a \left\{ \sum_{s',r} p(s',r|s,a) \left[r + \gamma v_\pi(s') \right] \right\}, \forall s$$

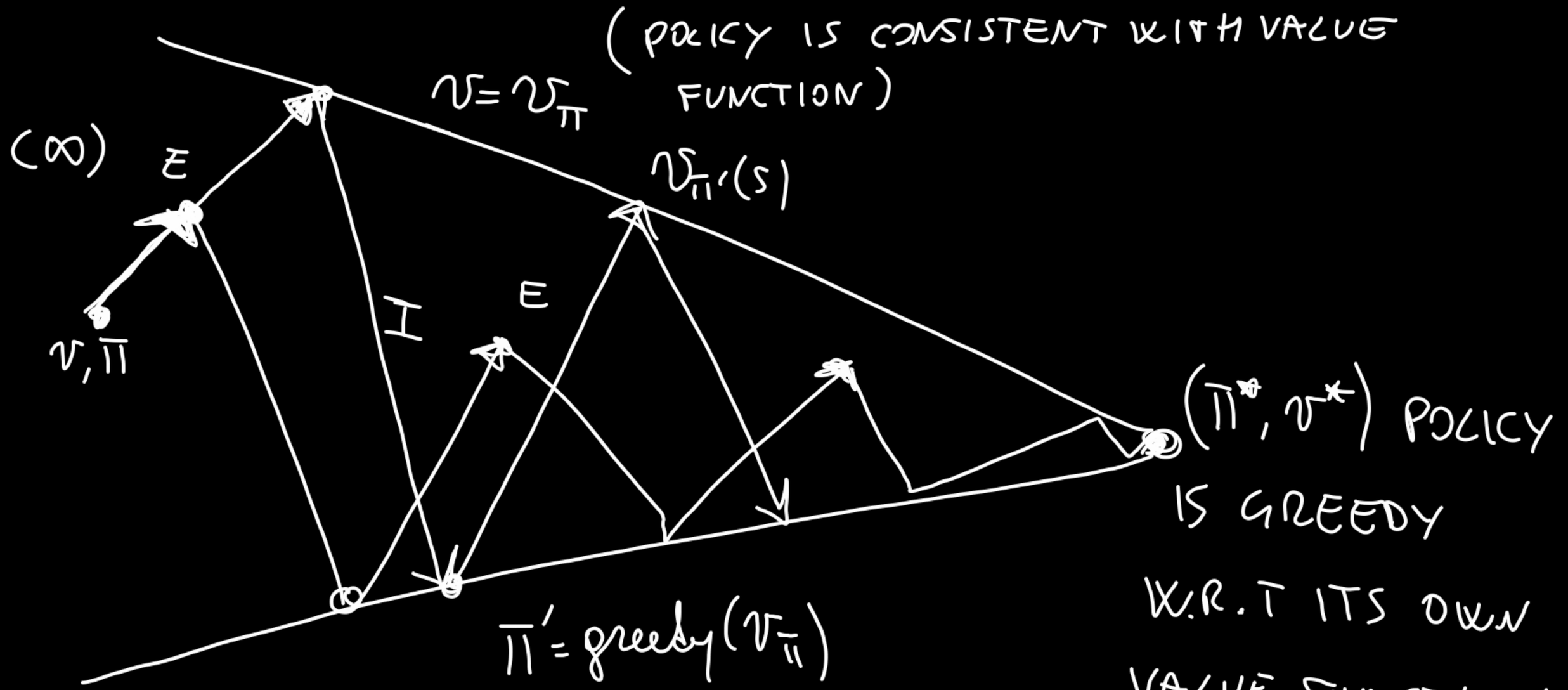


B.E. FOR THE OPTIMAL VALUE FUNCTION. π' OPTIMAL

$$\pi'(s) = \underset{a}{\operatorname{argmax}} \{ q_{\pi'}(s,a) \}, \quad \forall s$$

THE OPTIMAL POLICIES ARE GREEDY W.R.T THEIR OWN VALUE FUNCTION.

DIAGRAM TO REPRESENT POLICY ITERATION



IN ITERATIVE POLICY EVAL., WE USE BELLMAN EQ. AS THE UPDATE RULE.

VALUE ITERATION USES INSTEAD THE OPTIMAL BELLMAN EQ. AS UPDATE RULE.

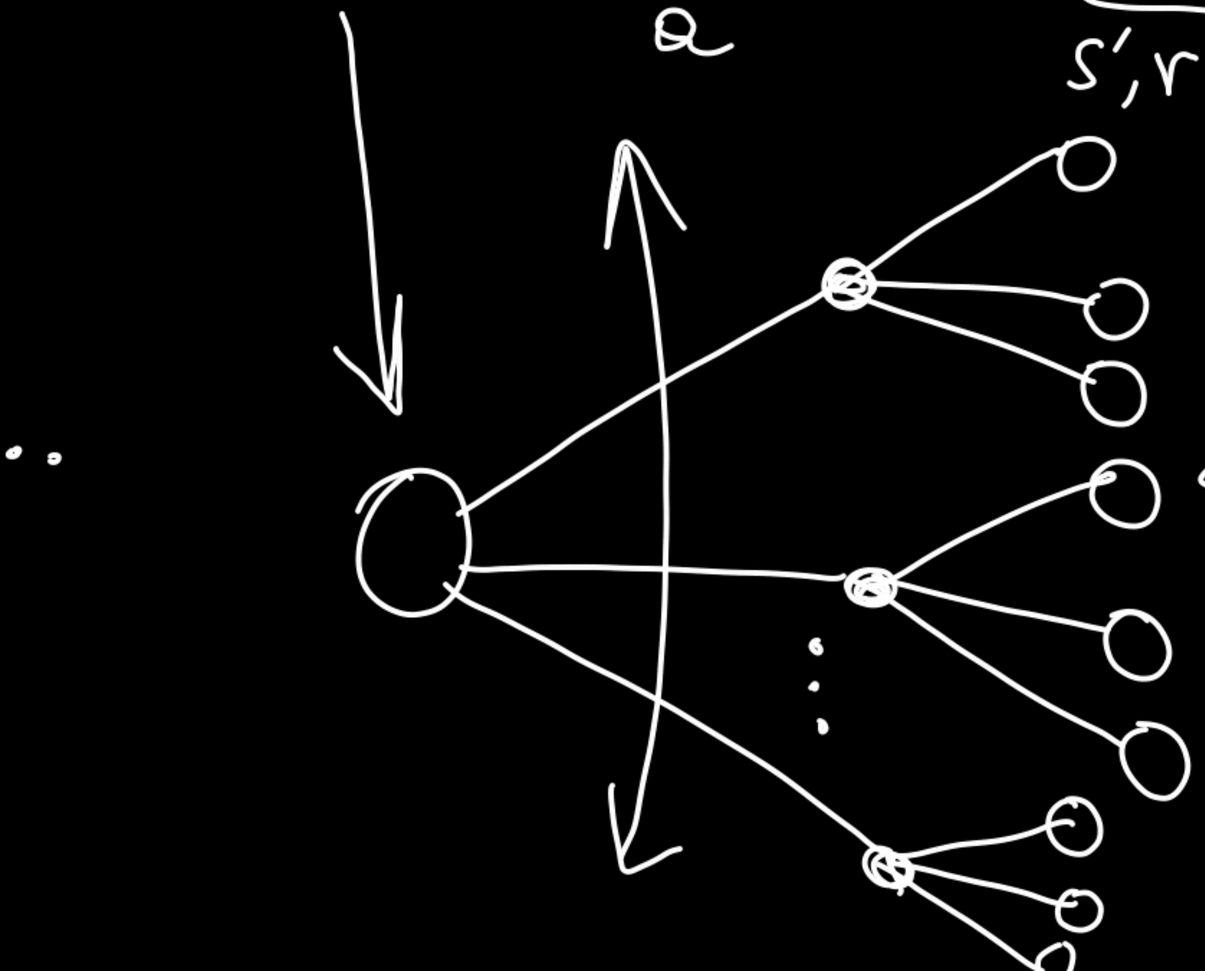
DYNAMIC PROGRAMMING METHODS ARE BASED ON

POLICY ITERATION: $\pi_0 \xrightarrow{E(\infty)} v_{\pi_0} \xrightarrow{I} \pi_1 \longrightarrow \cdots$

$\cdots \xrightarrow{E} v_{\pi_1} \xrightarrow{I} \pi_2 \cdots \pi^*$

EVALUATION IN DYNAMIC PROGRAMMING IS BASED ON EXPECTED UPDATES

$$v_\pi^{k+1}(s) = \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a)\left[r + \gamma v_\pi^k(s')\right], \forall s$$



MONTE CARLO RL METHODS — EPISODIC TASKS (□ TERM. STATE)

THEY WORK WITHOUT KNOWLEDGE OF MODEL OF THE ENVIRONMENT ($p(s', r | s, a)$).

KEEP SAME APPROACH BASED ON POLICY ITERATION.

- POLICY EVACUATION — PROBLEM. GIVEN A POLICY π, HOW CAN WE EVACUATE $v_\pi(s)$, WITHOUT p?

- POLICY IMPROVEMENT

$$\pi \downarrow \pi^*$$

$$s_1 \xrightarrow{?} v_\pi(s_1)$$

